

SHIVAM CHATTERJEE

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About

Aspiring AI/ML Engineer with hands-on experience in LLMs, RAG, Neural Networks, ML, Computer Vision, ETL pipelines and data visualization.

Education

Maulana Azad National Institute of Technology, Bhopal

2023-2027

Bachelor of Technology in Mechanical Engineering

CGPA - 8.6

Little Flower House, Varanasi, Uttar Pradesh

2021-2022

Class XII (PCM)

Percentage: 87.8

Skills

Technical Languages & Frameworks: Python, SQL, JavaScript, ReactJS

Data Science & AI: Machine Learning, RAG, LLMs, LangChain, Hugging Face, Sk-learn, Computer Vision

Analytics & Tools: Power BI, Excel, Pandas, DAX, Azure, Docker, Git, GitHub

Soft Skills: Leadership, Collaboration, Problem Solving, Communication, Adaptability

Personal Projects

ContextAI - RAG Document Assistant | *Python, LangChain, FAISS, Hugging Face, Ollama*

Demo | [🐙 GitHub](#)

- Developed a RAG-based document assistant enabling PDF upload and context-grounded question answering using local LLM inference.
- Implemented document processing pipeline with PDF extraction, text chunking, embeddings generation, and FAISS-based semantic retrieval.
- Deployed the backend API on Azure VM and hosted the frontend on Vercel, integrating cloud deployment with scalable document processing and retrieval workflows.

Real-Time Fruit & Vegetable Recognition using YOLOv8 | *Python, YOLOv8, OpenCV, Deep Learning*

[🐙 Github](#)

- Built a real-time object detection system using YOLOv8 for multi-class fruit and vegetable recognition from live webcam streams.
- Implemented bounding box localization, confidence thresholding, and class prediction pipelines for accurate real-time detection and visualization.
- Developed tomato ripeness estimation using RGB color analysis with OpenCV and NumPy for automated quality assessment.

Logistic Segmentation & Warehouse Assignment System | *Python, Pandas, Scikit-learn, ML*

[🐙 Github](#)

- Developed a logistics optimization system using K-Means clustering to segment customer demand zones based on geographical distribution.
- Implemented distance-based warehouse assignment to optimize delivery allocation and reduce operational travel distance.
- Evaluated clustering performance using Silhouette Score, achieving a score of 0.45 for cluster validation.

Work History

Toorynaad Society, MANIT Bhopal

2024 – Present

Lead Web Developer

[🌐 Tooryanaad.com](#)

- Led the web development team in designing and maintaining the Toorynaad Society platform, collaborating on feature development to enhance digital presence and improve event accessibility.

Banaras Locomotive Works (Indian Railways)

May 2025 – June 2025

Summer Industrial Trainee

Varanasi

- Gained practical exposure to locomotive manufacturing processes, production workflows, and quality control operations across multiple divisions.

Certifications

• **SQL and Relational Databases 101 – IBM**

[🌐 Certificate](#)